MNIST Classifier with Text-to-Speech Output

Objective: to use Google Text-to-Speech (gTTS) to translate the predicted digit for a particular test image into audible speech after training a neural network to classify MNIST handwritten digits.

Method:

1. **Dataset:** Loaded and preprocessed the standard MNIST dataset (normalized pixel values). 2. Model: Built a `keras.Sequential` model: `Flatten` -> `Dense` (128 units, ReLU) -> `Dropout` (0.2) -> `Dense` (10 units, Softmax).
2. **Training:** Compiled with 'adam' optimizer and 'sparse_categorical_crossentropy' loss, then trained for 10 epochs.
3. **Prediction:** Evaluated the model on the test set and made predictions. Selected the prediction for a specific test image (index 10).
4. **Text-to-Speech:** Installed the gTTS library. Converted the predicted digit (an integer) into a string. Used gTTS to generate an English MP3 audio file **("prediction_audio.mp3") of** the spoken digit.
5. **Output:** Displayed the test image, its predicted and true labels, and an audio player for the generated speech file.

Results:

- Model achieved high test accuracy: ~97.88%.
- Test loss was low: ~0.0744.
- The model correctly predicted the digit for the chosen test image (index 10) as '0'. * gTTS successfully generated an MP3 audio file speaking the predicted digit "0".
- The audio file was playable within the notebook environment.

Discussion: The experiment effectively illustrated a basic multimodal use case. High classification accuracy for MNIST digits was attained by a standard feedforward neural network. The gTTS library was then used to successfully translate the model's output—the predicted digit "0"—into spoken audio. This illustrates how various AI/ML components can be combined by showing a simple pipeline from visual input (image) to classification (numerical prediction) to speech output (auditory).